

Vinh Pham

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Education

Georgia Institute of Technology

Atlanta, GA

B.S. in Computer Science | Honors Program Scholar | **GPA: 4.0/4.0**

May 2028

- **Coursework:** Machine Learning, Artificial Intelligence, Algorithms Honors, Perception & Robotics, Systems & Networks, Computer Organization & Programming, Data Structures & Algorithms, Discrete Math, Linear Algebra, Multivariable Calculus
- **Languages:** Proficient in **Python, Java** | Familiar with C, C#, Assembly, SQL, JavaScript, MATLAB
- **Frameworks:** PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, SciPy | Django, React, Next.js, .NET 8
- **Tools:** AWS (Bedrock/AgentCore), Terraform, Git, Docker, PostgreSQL, MongoDB, REST APIs, Google Cloud Platform, Redis

Work Experience

Cox Automotive

Austin, TX

Software Engineer Intern | Python, C#, Terraform, AWS, SQL, Docker, Redis

May 2026 - Present

- Building a RAG-powered Slack agent with AWS AgentCore that answers engineer and support ticket queries from an auto-synced, multi-sourced knowledge base—achieving **97% answer accuracy** and deflecting **~40%** of **8 hrs/wk/engineer** support overhead
- Led end-to-end modernization of the Overwatch platform's architecture and CI/CD, designing a parallelized two-pass search-tree retrieval pipeline with Redis caching, reducing latency of database query and bulk edit of **122,000+ client subscriptions by 85%+**

Cortado Group

Remote

Full Stack Software Engineer Intern | Python, Django, React, PostgreSQL

January 2026 - May 2026

- Architected a **PostgreSQL-backed Django project** to convert transcripts into GTM growth insights for PE portfolio firms
- Developed a dual-path AI scoring engine and Slack-synced agents to automate bottleneck detection for C-suite-led deployment

Dell Technologies

Austin, TX

Software Engineer Intern | Python, C, OpenAI, Whisper

February 2025 - May 2025

- Developed an AI-powered robot that translates speech into Mermaid diagrams, facilitating concept visualization and discussion
- Engineered dual-LLM prompts to automate diagram generation and consistent G-code syntax for precise, reliable motor control
- Designed a **multithreaded-RAG pipeline** to **optimize runtime by 80%** and enable real-time deployment for Dell executives

Research Experience

Mind, Brain, Education (MBE) Lab - Georgia Institute of Technology

Atlanta, GA

Undergraduate Researcher | Principal Investigator: Dr. Maysam Nezafati

January 2026 - Present

- Engineering and optimizing an fNIRS preprocessing pipeline for the ASTaUND dataset that utilizes **attention-based LSTM autoencoders** to identify neural biomarkers of motor skill acquisition and improve surgical expertise classification accuracy
- Developing a novel **multivariate delay symbolic transfer entropy (TE) framework** to quantify non-linear information flow and establish a robust connectivity model used to design data-driven GT BME curricula and adaptive surgical training protocols

Applied Research Laboratories (ARL) - University of Texas at Austin

Austin, TX

Undergraduate Researcher | Principal Investigator: Dr. Christina Naify

June 2025 - August 2025

- Designed an experimental study for evaluating dynamic variability in 180+ additively manufactured beams and automated laser Doppler vibrometer tuning via MATLAB to achieve **95%+ precision** – a **30% improvement** over previous research methods
- Optimized regression algorithm **runtime by 50%**, accelerating statistical analysis to drive research conclusions and publication

Projects

[SmartPathAI](#)

Atlanta, GA

Project Lead - Big Data Big Impact | Python, React, FastAPI, Node.js, Neo4j, GCP Cloud Run

August 2025 - Present

- Led deployment of an AI learning platform for **20k+ GT students**, converting PDFs into interactive Neo4j knowledge-graph study paths; coordinated architecture across **5 teams** and partnered with GT College of Computing for faculty material integration
- Designed multi-modal PDF pipeline with pdfplumber, AI OCR, and LLM-based image processing to extract concepts into nodes
- Built GraphRAG pipeline with OpenAI embeddings, parallel extraction, and JSON-sanitized Neo4j upsert, **~50x** faster generation

AI-Powered Prosthetic Arm

Atlanta, GA

AI/ML Engineer - GT Medical Robotics Club, Team Limbo | Python, OpenCV, PyTorch, Streamz

August 2025 - February 2026

- Designed an **unsupervised autoencoder pipeline** mapping live EMG signals to 6D gesture vectors for prosthetic arm control
- Developed computer vision gesture tracking dashboard via OpenCV/Mediapipe, to offer ground truth labels for model training

Leadership

[Project 1600](#)

Remote

Founder

May 2023 - January 2026

- Established a multi-district SAT mentorship program and curriculum, elevating the mean scores of **15+ students by 160+ points**
- Designed an automated website for enrollment management, curriculum planning, and tutor recruitment, generating **320+ visits**